GLOBAL INDUSTRY CHALLENGE 2026

TEAM COVER PAGE

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Challenge Name: Quantum-Enhanced Strategic Siting of Energy Storage and Microgrids

Phase #: 3

Team Name: Entangled Trio

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1 Focus Area, Rationale, and Stakeholder Relevance

This project addresses the strategic siting and capacity sizing of clean energy microgrids and storage systems within power distribution grids. Siting decisions represent a high-stakes, combinatorial planning challenge under complex electrical constraints. To guide real planning outcomes, we model a multi-objective optimization problem that co-optimizes grid resilience (C_{R}), capital installation cost (C_{I}), and operational voltage control (C_{VC}). Our Jupyter notebook on siting optimization, AI load scaling and $N-1$ resilience are available at [1].

The primary stakeholders and beneficiaries of this methodology include:

- **Electric Utilities and ISOs:** Enables optimal capital allocation for grid assets and helps maintain power quality under dynamic load conditions.
- **Department of Energy (DOE):** Accelerates grid decarbonization and modernization mandates through quantum-enhanced optimization frameworks.
- **Local Communities:** Enhances energy security and resilience against catastrophic outages during extreme weather events.

2 Technical Methodology and Hybrid Quantum Integration

We run a hybrid Benders-QAMOO workflow on the IEEE-33 distribution bus network, combining Quantum Approximate Multi-Objective Optimization (QAMOO) [2] with classical Benders decomposition. Inspired by [3] but with a different architecture partition, the optimization is divided into a classical Master Problem (MP) and an AC power-flow Subproblem (SP). The siting problem is formulated as a mixed-integer linear program (MILP) encoded in a Qiskit `QuadraticProgram` [4], co-optimizing the three objectives from above under the scalarized minimization: $\min_{\overline{x}\in\{0,1\}^N} \quad C(\overline{x}) = \lambda_{\mathrm{I}}C_{\mathrm{I}} + \lambda_{\mathrm{R}}C_{\mathrm{R}} - \lambda_{\mathrm{VC}}C_{\mathrm{VC}}$

2.1 Grid Preprocessing: Objective Coefficient Computation on the AI-Shocked Grid

Before the Benders-QAMOO siting optimization runs, the pipeline computes three per-bus objective coefficients on an AI load-stressed version of the IEEE-33 grid. The resulting coefficients (V_n, r_n, $dist_n$) directly parameterize the QUBO diagonal terms for each candidate bus.

AI load shock injection. Total demand growth (ΔP_{DC}) is sourced from the PNNL/EPRI TELL dataset [5] under the high-growth scenario for the TVA balancing authority, capped at 25% of baseline system load to ensure AC power flow convergence on the IEEE-33 test grid. Load is injected exclusively onto *hub buses*—the top 15% of load buses ranked by graph degree centrality—in proportion to each bus's existing demand. The empirical basis for this geographic concentration model is documented in Section 3.

Voltage sensitivity V_n. On the AI-shocked grid, a base AC power flow is solved using a cascading solver fallback (Newton–Raphson $\rightarrow$ Iwamoto damped NR $\rightarrow$ backward/forward sweep). For each candidate bus, reactive power probes of $\pm\Delta Q = 5$ MVAR are injected and the reduction in the squared voltage deviation score $\|\mathbf{V}-\mathbf{1}\|^2$ is recorded. The bus-wise maximum improvement, normalized to $[0,1]$, gives V_n: higher values indicate buses where reactive support most effectively corrects voltage violations.

Investment cost proxy r_n. Existing bus load (MW) serves as a proxy for installation cost and civil-works complexity. Buses with zero load are assigned a floor of 10% of the mean network load. The resulting values are normalized to

[0, 1].

Contingency-weighted resilience $dist_n$**.** Rather than an exhaustive outage sweep, the pipeline uses an $N-1$ strategy grounded in empirical hazard probabilities from the PNNL EAGLE-I × DOE-417 dataset [6] (tabulated in Section 3). Each transmission line is assigned to one of three hazard categories (weather, physical/security, equipment/operational) based on thermal loading and hub-bus connectivity percentile thresholds. The highest-loaded line within each category is taken out of service, and per-bus voltage recovery under $\pm\Delta Q$ probes is measured. The final resilience metric is the empirically weighted average: $dist_n = \sum_{\text{cat}} P(\text{cat}) \cdot \hat{R}_n^{\text{cat}}$ where $\hat{R}_n^{\text{cat}}$ is the normalized per-bus voltage recovery improvement under the category-representative $N-1$ outage.

QUBO parameterization. These three coefficients are used to parameterize the self-loop diagonals of the three Ising objective graphs, encoding the multi-objective siting problem into the QAMOO quantum layer: $Q_{i,i} = -\alpha V_n[i] + \mu r_n[i] - \nu\, dist_n[i] + \lambda(1-2K)$ where α, μ, ν are scalarization weights, K is the budget constraint (number of assets to place), and $\lambda = 10.0$ is the Benders feasibility penalty from the classical preprocessing step.

2.2 Benders-QAMOO Workflow

The MP handles the discrete layout coordinates, selecting the binary portfolio of bus locations for asset placement ($\hat{x}^{(t)} \in \{0,1\}^N$). It is solved using CPLEX [7]. The candidate layout $\hat{x}^{(t)}$ is passed to the SP evaluator, which runs an AC power-flow simulation using pandapower [8].

To embed feasibility constraints into the QAMOO quantum layer, we convert the collected cuts into quadratic penalty terms $p_i = \lambda \cdot \hat{x}_i^{(t)}$. Once the penalized graphs are compiled, we train a multi-objective warm-start QAOA on a scalarized representation of the three objectives. The Pareto front is evaluated by executing batched sample runs of size $S = 10$ using Dirichlet-distributed scalarization weights.

3 Data Sources and Modeling Assumptions

This section documents the provenance of all external datasets used in the Phase 3 pipeline, explicitly delineating what is derived from real-world government data versus what constitutes a modeling assumption.

3.1 Data Source Provenance

3.2 AI Data Center Load Injection Model

What is from real data. The total megawatt addition for the TVA Balancing Authority under the high-growth scenario ($\Delta P_{\text{DC}} = 383.40$ MW) is sourced directly from the PNNL/EPRI TELL dataset [5]. This dataset provides hourly Balancing Authority–level load projections with and without data center demand under moderate and high growth scenarios, covering all 66 U.S. BAs.

What is a modeling assumption. The TELL dataset operates at the BA level and does *not* specify which substations within a BA absorb data center demand. Since IEEE benchmark grids are synthetic topologies with no geographic mapping, any sub-BA bus-level allocation is necessarily a modeling choice.

We adopt a **sparse hub-bus injection model** based on the following empirical evidence of geographic clustering:

Table 1: External Dataset Provenance

Pipeline Input	Source Dataset	Provenance
Total AI load addition (ΔP_{DC})	PNNL/EPRI IM3+EPRI Data Center Load Projections (TELL BA)	Real govt. data
Outage event category probabilities $P(\mathrm{cat})$	PNNL Event-Correlated Outage Dataset (EAGLE-I × DOE-417, 2014–2023)	Real govt. data
Bus-level AI load allocation (which buses receive ΔP_{DC})	Degree-centrality hub selection with 15% concentration ratio	Modeling assumption
Line hazard categorization (Weather / Physical / Equipment)	Thermal loading and degree-centrality percentile thresholds	Modeling assumption
IEEE bus test systems (14/33/39/118)	pandapower standard library [8]	Synthetic benchmark

- **PJM (2025):** Data centers account for 94% of projected load growth, with demand concentrated in the Northern Virginia corridor (“Data Center Alley”) [9].
- **IEA (2025):** Approximately 50% of U.S. data centers under development cluster near existing large facilities at high-voltage interconnection points [10].
- **LBNL Queued Up (2025):** Approximately 80% of large-load interconnection requests target fewer than 15% of available substations [11].
- **WECC (2024):** Data center interconnection requests are geographically concentrated at a small number of high-capacity substations [12].

Based on these sources, we set $f_{\mathrm{hub}} = 0.15$ (15% of load buses) as the concentration fraction. This value is consistent with the LBNL finding that <15% of substations absorb the majority of large-load requests. The selected hub buses are the top-$\lceil f_{\mathrm{hub}} \cdot N_{\mathrm{load}} \rceil$ buses ranked by graph degree centrality (a proxy for high-voltage interconnection capacity). Within the selected hub set, demand is distributed proportionally to existing bus load:

$$P_n^{\mathrm{shocked}} = P_n^{\mathrm{base}} + \frac{P_n^{\mathrm{base}}}{\sum_{k \in \mathcal{H}} P_k^{\mathrm{base}}} \cdot \Delta P_{\mathrm{DC}}, \quad n \in \mathcal{H} \tag{1}$$

where $\mathcal{H}$ is the set of hub buses and $P_n^{\mathrm{base}} = 0$ for $n \notin \mathcal{H}$ (no injection at non-hub buses).

To ensure AC power flow convergence on small test grids (IEEE-14, IEEE-33), we additionally cap the total injection at 25% of baseline system capacity: $\Delta P_{\mathrm{DC}}^{\mathrm{eff}} = \min(\Delta P_{\mathrm{DC}}, 0.25 \cdot \sum_k P_k^{\mathrm{base}})$.

3.3 $N - 1$ Contingency Resilience Model

What is from real data. The empirical event-category probabilities used to weight the $N - 1$ contingency metric ($dist_n$) are derived from the PNNL Event-Correlated Outage Dataset [6], which synthesizes 10 years (2014–2023) of EAGLE-I county-level outage records with DOE-417 Electric Emergency Incident and Disturbance Reports, covering 663 major grid disturbance events across the continental United States. The resulting empirical probabilities are:

What is a modeling assumption. The mapping from hazard categories to specific transmission lines within IEEE benchmark grids is a modeling assumption, since these synthetic topologies have no geographic or physical-asset

Table 2: Empirical Outage Category Probabilities (OEDI 6458, 2014–2023)

Event Category	$P($**cat**$)$
Weather / Natural Disaster	44.9%
Physical / Security	27.3%
Equipment / Operational	24.6%
Other / Unknown	3.0%
Cyber	0.2%

metadata. We use a **physics-grounded percentile categorization** scheme that generalizes across any grid size:

- **Weather hazards** → Lines in the top-20% thermal loading percentile (proxy: exposed, high-stress corridors).
- **Physical/Security hazards** → Lines touching hub buses in the top-15% degree centrality percentile (proxy: high-value interconnection targets).
- **Equipment/Operational hazards** → Remaining lines (proxy: routine failure modes).

For each category, a single representative line (the highest thermally loaded within its category) is taken out of service ($N-1$ contingency), and reactive power probes (±5.0 MVAR) at each candidate bus measure voltage recovery improvement (R_n^{cat}). The final per-bus resilience metric is:

$$dist_n = \sum_{\text{cat}} P(\text{cat}) \cdot \frac{R_n^{\text{cat}}}{\max_k R_k^{\text{cat}}} \tag{2}$$

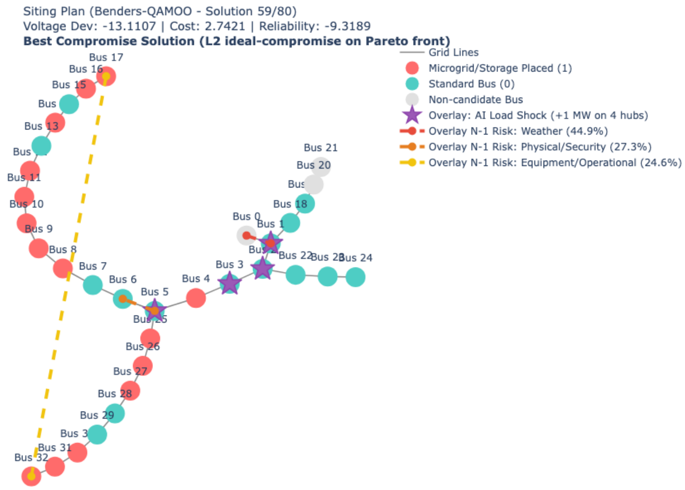
3.4 AC Power Flow Solver Robustness

Under stressed loading conditions (AI load shock + $N-1$ outage), the standard Newton-Raphson (NR) AC power flow solver may fail to converge. We implement a cascading fallback strategy: NR (max_iter = 100) → Iwamoto Damped NR → Backward/Forward Sweep (BFSW). This ensures convergence across all test cases without masking genuine physical infeasibility.

4 Experimental Results and Discussion

Figure 1 translates the abstract Pareto metrics into a concrete siting recommendation: the best-compromise solution selected from the Benders-QAMOO Pareto front (L2 norm to the ideal point) is overlaid directly onto the IEEE-33 distribution bus topology. Red nodes are buses selected for microgrid or storage placement, teal nodes are unselected candidate buses, and grey nodes are non-candidate buses. Star markers denote AI data-center load shock injection hubs, and dashed colored edges highlight the representative N-1 contingency lines per outage-hazard category.

Sample efficiency and convergence. Beyond final Pareto quality, Benders-QAMOO exhibits a decisive *convergence rate* advantage over native QAMOO. Figure 2 traces hypervolume as a function of samples evaluated during the siting optimization. Benders-QAMOO reaches a hypervolume of ~330 in under 12,500 samples, a threshold that native QAMOO does not approach even after 40,000 samples — a roughly 3× improvement in sample efficiency. This accelerated convergence is a direct consequence of the Benders decomposition: by separating the integer master problem from the continuous sub-problems, the optimizer concentrates quantum circuit evaluations on the most informative regions of the search space rather than performing undirected exploration.



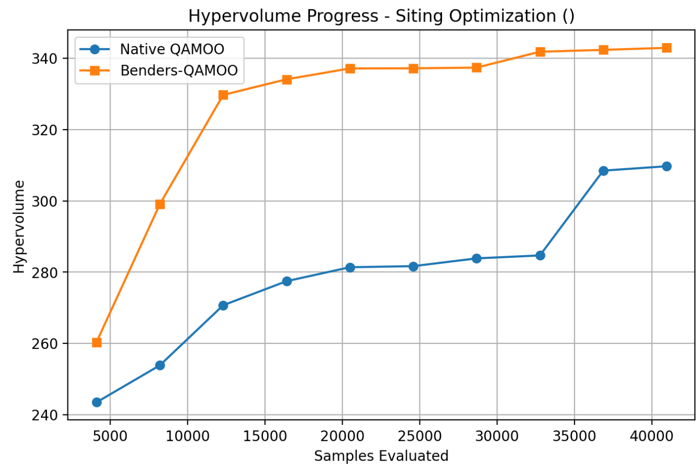


Figure 1: Best-compromise microgrid and storage siting plan on the IEEE-33 distribution bus grid, selected from the Benders-QAMOO Pareto front. **Red:** bus selected for microgrid/storage placement; **teal:** candidate bus not selected; **grey:** non-candidate bus (slack). Star markers indicate AI data-center load shock injection hubs; dashed edges mark the representative N-1 contingency line per hazard category (weather, physical/security, equipment).

Figure 2: Hypervolume progress vs. samples evaluated for Benders-QAMOO (orange) and native QAMOO (blue) on the siting optimization task. Benders-QAMOO converges to near-plateau hypervolume in roughly one-third the samples required by native QAMOO, demonstrating superior sample efficiency.

5 Gap Analysis

- **Error Mitigation:** We were not able to test on IBM QPUs and thus had trouble with the Samplomatic error-mitigated pipeline which uses the Executor primitive that requires IBM Heron QPU. We have a workaround with Qiskit Aer simulator, but it is not supported by Samplomatic and thus not recommended due to poor performance.
- **Larger Datasets:** Our workflow was designed for large datasets like IEEE-118, but we only tested on IEEE-33 due to compute limitations. Our AI data-center shock and $N-1$ contingency hazards are modeling assumptions that would shine in real-world scale.
- **cuTensorNet acceleration:** Our code automatically leverages cuTensorNet for MPS calculations, however we were compute constrained for this challenge and could not test on the larger datasets that would truly show the benefits of this approach.
- **Quantum Advantage:** Practical quantum advantage for optimization is an open problem, and we will continue to monitor the progress in quantum algorithms and hardware.

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